REPORTING PERIOD: 7 MARCH 2024 TO 20 MARCH 2024

Geographic Data Science Lab

UNIVERSITY OF LIVERPOOL

SITUATION REPORT: FLASH FLOODS, WEST SUMATRA | INDONESIA
DIGITAL TRACE DATA | EXPERIMENTAL

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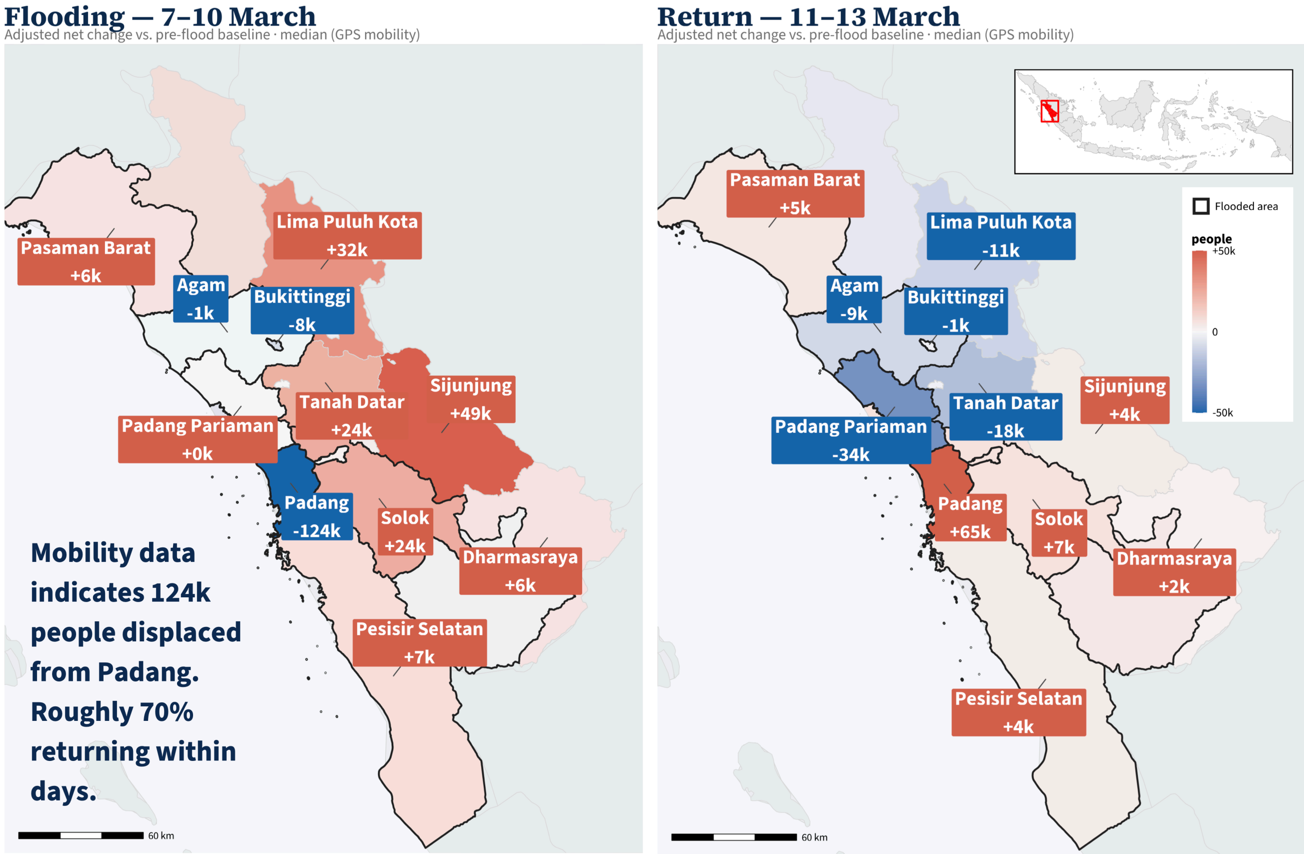

Figure 1. Estimated population movement analysis, West Sumatra, 07–13 March 2024.

Event Trigger

- Extreme rapid-onset rainfall: **~300 mm in 6 hours** late on 7 March, triggering **flash floods** and **landslides** across mountainous West Sumatra.
- **No advance warning issued** — only general monsoon forecasts from BMKG.

ESTIMATED DISPLACED - 07–20 MAR 2024

132,000

people displaced · GPS mobility data

16 deaths · **128,727** displaced · IFRC & PMI, 2024 [1]

Insights

This situation report uses **digital trace data** as an alternative way to measure population dynamics in near real-time over the reporting period, applying a GPS-phone approach [2].

Movements are measured as the **deviation from usual mobility** — a design that isolates the change attributable to the floods. Key advantages are **timeliness** and **spatial granularity**.

- **Padang City** epicentre: GPS estimates **~100,000 displaced on 8 March**, with rapid return within days
- **Padang Pariaman and Agam** show delayed displacement, reflecting slower inland flood progression
- **Pesisir Selatan** displacement only observed March8th; GPS signal consistent with BNPB reports; spillover displacement observed in surrounding areas
- **Solok, Limapuluh Kota, Pasaman and West Pasaman** reported affected by humanitarian sources; GPS signals indicate more limited displacement
- Return movement underway by **11 March** across affected kabupaten (admin 2)

Situation Report

West Sumatra, Indonesia: Floods — March 2024

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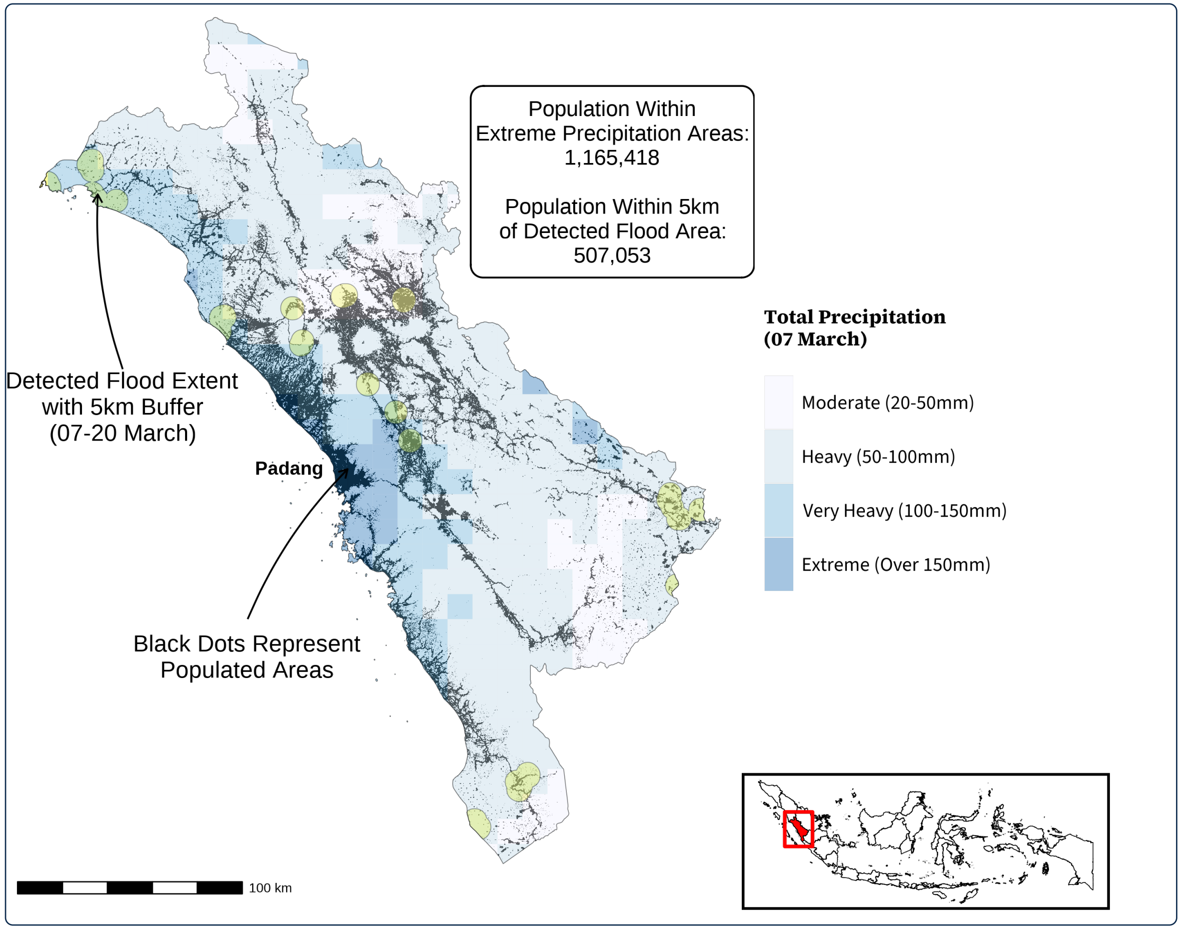

Figure 2. Rainfall and flood patterns, West Sumatra, 07–20 March 2024.

- *Population: WorldPop*
- *Rainfall: NASA Global Precipitation Measurement (GPM) IMERG Late Run*
- *Flood: NASA Land, Atmosphere Near real-time Capability for Earth observation (LANCE), MODIS satellite*

Flood context

- The heaviest rainfall was concentrated around the province's capital, **Padang**, home to approximately **1 million residents** and the hub of the region's road network.
- Satellite data captured some flood waters, though heavy cloud cover limited coverage — Figure 2 **does not show the full extent of flooding in the region.**
- Approximately **1.2 million people** live in the high-rainfall area and **507,000** within 5 km of a detected flood, in the two weeks since the event was declared.

Warning gap

- **No effective flash-flood warning system** was in place in the affected districts prior to the event — only general monsoon-season forecasts from BMKG [3].
- The speed of onset (**300 mm in 6 hours**) meant effective lead time was **zero**.
- Early warning systems at **23 sites** in Agam, Tanah Datar and Padang Panjang were **planned and installed after the disaster** (May 2024), confirming the pre-event gap.

The absence of advance warning makes digital trace data a critical alternative for understanding population response in near real time.

Limitations

- **Within-city movement:** GPS data suggest population movement within Padang itself, visible in the fine-grained spatial signal but not reflected in this report.
- **Landslide impacts:** communities affected by landslides are difficult to assess, given the lack of detailed landslide data.

Patterns were assessed and validated against humanitarian reporting (Save the Children, 2024). *Readers are strongly encouraged to consult the caveats before interpreting these findings.*

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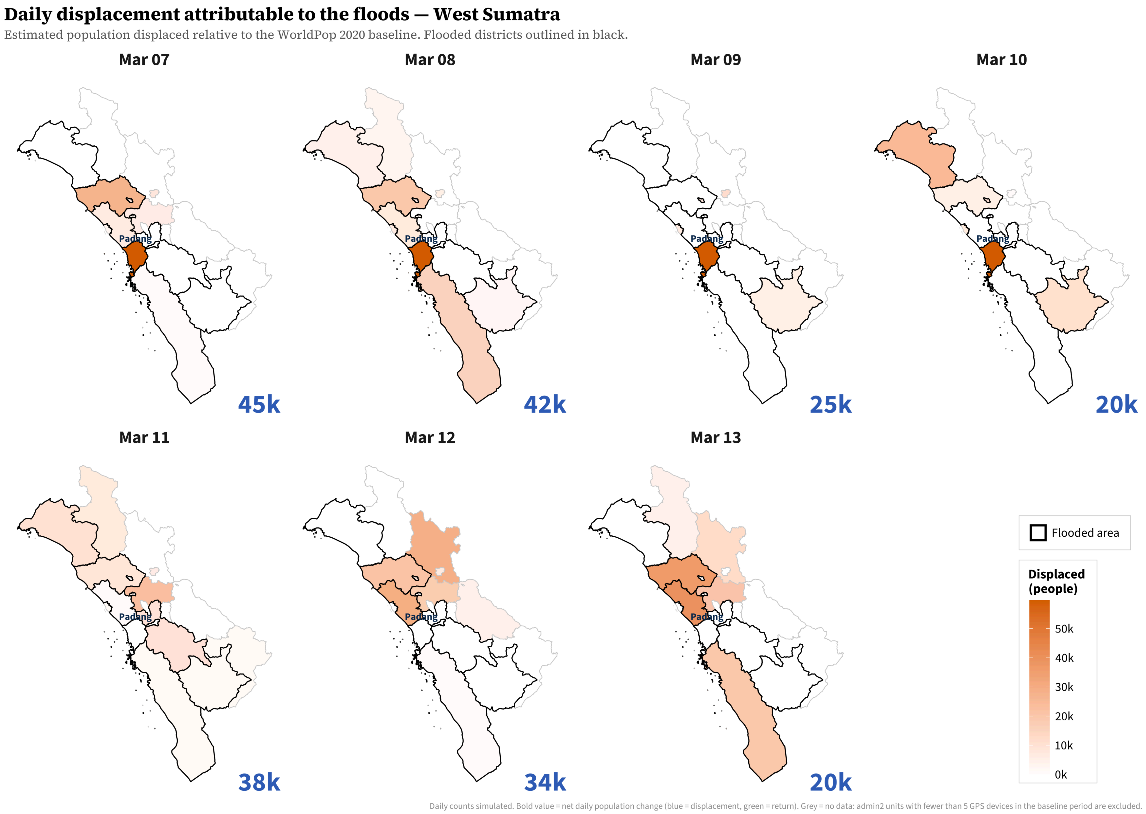

Figure 3. Estimated daily displacement, West Sumatra, March 2024.

Data Sources

GPS mobility data. Device-level GPS locations from mobile phones in West Sumatra, from commercial vendor aggregating multiple apps to reduce single-source bias. Device counts scaled to population using WorldPop 2020.

Administrative boundaries. OCHA Humanitarian Data Exchange (HDX).

Flood extent. [IFRC | MODIS | GroundSource |.]

GPS data are not a representative sample; figures exclude populations with limited device access and are indicative of patterns, not precise counts.

Methodology

Step 1. Baseline mobility from pre-flood GPS data (1–6 March 2024).

Step 2. Flooded areas are compared with dry ones to separate flood-driven movement from normal change (7–20 March 2024) (DiD).

Step 3. 1-day displacement threshold; sensitivity checks at 3 and 7 days.

Indicator	Available	Use in this report
GPS / mobile device data	YES	Primary proxy for pop change
MODIS satellite flood data	YES	Flood extent delineation
Copernicus GloFAS	PARTIAL	Intermittent coverage; not used
Cloudflare HTTPS requests	YES	Not used in this report
Google search activity	PARTIAL	Not used in this report
Wikipedia pageviews	PARTIAL	Not used in this report
Night-time lights	PARTIAL	Not used in this report
Meta population estimates	NO	Not used in this report

Table 1. Digital trace data available for Indonesia. Sources marked PARTIAL are available but with limited geographic or temporal coverage or resolution.

GPS mobility data
Device locations
WorldPop 2020
baseline population

→

Baseline model
1–6 Mar 2024

→

Post-flood estimation
7–20 Mar 2024

→

DiD adjustment
flooded vs. unflooded units

→

Displacement estimates
Population movement index

Bibliography

[1] IFRC and Palang Merah Indonesia, “Indonesia: Flood – 03-2024 – Floodings across West Sumatera #1.” [Online]. Available: https://reliefweb.int/report/indonesia/indonesia-flood-03-2024-floodings-across-west-sumatera-1-2024-03-11

[2] R. Iradukunda, F. Rowe, and E. Pietrostefani, “Estimating internal displacement in Ukraine from high-frequency GPS mobile phone data,” *Humanities and Social Sciences Communications*, vol. 12, no. 1, p. 1863, 2025.

[3] Badan Meteorologi, Klimatologi, dan Geofisika (BMKG), “Badan Meteorologi, Klimatologi, dan Geofisika.” [Online]. Available: https://www.bmkg.go.id/